Transfer Learning:
ResNet-50 vs. DenseNet-121 on CalTech-101

1. Introduction

Pre-trained deep neural networks are widely used for new tasks, but deciding how to adapt them effectively remains non-trivial. Transfer learning addresses this by reusing representations learned on large datasets such as ImageNet. A common approach is fine-tuning, where selected layers are updated for a target task. However, standard fine-tuning treats all parameters within chosen layers as equally relevant, which may not hold in practice, particularly on small datasets like CalTech-101, where over-parameterisation risks overfitting.

This report evaluates both standard fine-tuning and an adaptive filter-importance strategy on CalTech-101 using two architectures: ResNet-50, which uses additive residual connections

$H(x) = F(x) + x$, and DenseNet-121, which uses dense feature concatenation

$x_l = H([x_0,...,x_{l-1}])$. The goal is to analyse how these architectural differences affect accuracy, convergence, and parameter efficiency across fine-tuning strategies.

2. Methodology

2.1 Dataset and Preprocessing

The CalTech-101 dataset contains 9,144 RGB images across 102 classes (including BACKGROUND_Google), with notable class imbalance. The dataset was partitioned into 70% training (6,400 images), 10% validation (914), and 20% test (1,830) using a fixed random seed (42) for reproducibility. Images were resized to 256×256, centre-cropped to 224×224, and normalised using ImageNet channel-wise statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]) to ensure compatibility with pretrained weights. Random horizontal flipping and colour jitter were applied as standard training augmentation. Batch size was set to 32, a practical balance between memory efficiency and stable gradient estimates on the available GPU (NVIDIA RTX 4000 Ada).

2.2 Model Architecture and Head Modification

ResNet-50 uses residual connections $H(x) = F(x) + x$, enabling stable gradient flow in deep networks. It produces a 2048-dimensional feature vector with approximately 23.7M parameters and ~4.1 GFLOPs per forward pass.

DenseNet-121 uses dense connectivity where each layer receives concatenated feature maps from all preceding layers. This promotes explicit feature reuse and yields a more compact model (~7.06M parameters, ~2.9 GFLOPs) with a 1024-dimensional output.

For both models, the original 1000-class classification head was replaced with a fully connected layer mapping to 102 classes (2048→102 for ResNet-50; 1024→102 for DenseNet-121). All pretrained backbone weights were retained from ImageNet initialisation.

2.3 Training Strategies

Strategy I Standard Fine-tuning: The backbone was frozen and only higher-level layer groups were unfrozen. Two configurations were evaluated for each model: (i) top-2 layers, and (ii) top-3 layers. For ResNet-50, top-2 unfreezes fc + layer4; top-3 additionally unfreezes layer3. For DenseNet-121, top-2 unfreezes classifier + denseblock4; top-3 additionally includes transition3. Lower layers were kept frozen to preserve general ImageNet features (edges, textures) while adapting task-specific representations.

Strategy II Adaptive Fine-tuning (Filter Importance Ranking): The frozen model was first evaluated over the full training set to compute a per-filter importance score defined as the average activation magnitude: score(f) =

mean(|output_f|) over all batches and spatial positions. Two variants were implemented: (i) Last-block filtering importance computed only within the final convolutional block (layer4 for ResNet-50; denseblock4 for DenseNet-121), with the top 25% of filters selected for updating via gradient masking; (ii) Global filtering importance computed across all convolutional layers globally, with the top 25% selected globally. In both cases, unselected filters receive zero gradient during backpropagation.

2.4 Hyperparameters

Hyperparameters were selected via a grid search over optimizer (SGD, Adam), learning rate (1e-3, 1e-4, 1e-5), and scheduler (StepLR, ReduceLROnPlateau), each evaluated for 5 epochs. Adam consistently outperformed SGD when only a subset of layers is trainable, due to its per-parameter adaptive updates. The selected configurations were: ResNet-50 Adam, lr=1e-4, StepLR; DenseNet-121 Adam, lr=1e-3, StepLR. A lower learning rate was used for ResNet-50 given its larger parameter space, while DenseNet-121's compact structure supports a higher rate. Weight decay of 1e-4 was applied throughout as L2 regularisation. StepLR decayed the learning rate by a factor of 0.1 every 10 epochs. Early stopping with patience 5 was applied on validation loss to prevent overfitting, with the best checkpoint selected by validation accuracy. The same best configs were applied to both cases (top-2 and top-3) of each model.

3. Results

3.1 Standard Fine-tuning

Metric	ResNet C1	ResNet C2	DenseNet C1	DenseNet C2
Unfrozen Layers	fc + layer4	fc+layer4+layer 3	cls+db4	cls+db4+tr3
Total Params	23.7M	23.7M	7.06M	7.06M
Trainable Params	15.2M (64%)	22.3M (93.9%)	2.26M (32.1%)	2.79M (39.5%)
GFLOPs	4.1	4.1	2.9	2.9
Best Val Acc (%)	94.64	93.54	93.98	94.31
Test Acc (%)	93.28	92.51	90.93	93.61
Macro F1	0.8991	0.8828	0.8678	0.8991
Train Time (s)	100.9	147.1	100.1	333.0
Time/Epoch (s)	11.2	14.7	11.1	11.5

Table 1: Standard Fine-tuning All 4 Cases

ResNet-50 Case 1 (fc + layer4) achieved the best test accuracy among ResNet configurations at 93.28% with F1 of 0.8991 in only 100.9s. Case 2 (adding layer3) showed clear overfitting validation loss rose steadily after epoch 3 while train loss continued to fall, and test accuracy dropped to 92.51%. Unfreezing layer3 increased trainable parameters from 15.2M to 22.3M, exceeding what 6,400 training images can support without overfitting.

DenseNet-121 Case 2 (cls + denseblock4 + transition3) achieved the best overall test accuracy at 93.61% with validation loss plateauing stably at ~0.20 for 28 epochs the most stable convergence observed. Case 1 (cls + denseblock4 only) was the weakest result at 90.93% with noisy, unstable training. Without transition3, denseblock4 receives frozen upstream features unadapted to CalTech-101, creating an information bottleneck.

3.2 Adaptive Fine-tuning

Metric	ResNet-50	DenseNet-121
Method	Last-Block (layer4)	Last-Block (denseblock4)
Trainable Params	15.2M (63.9%)	1.87M (26.4%)
Best Val Acc (%)	91.47	94.20
Test Acc (%)	91.26	92.68
Macro F1	0.8431	0.8952
Train Time (s)	114.4	287.0
Global Adaptive Finetuning		
Trainable Params	22.4M (94.3%)	5.54M (78.5%)
Best Val Acc (%)	85.89	86.43
Test Acc (%)	85.52	84.59
Macro F1	0.7779	0.7677
Train Time (s)	256.3	666.8

Table 2: Adaptive Fine-tuning Last-Block and Global variants

Last-block adaptive finetuning showed strongly divergent behaviour between the two architectures. DenseNet-121 achieved 92.68% test accuracy with only 26.4% trainable parameters (1.87M) and stable convergence over 25 epochs the most parameter-efficient result overall. ResNet-50 exhibited textbook overfitting: validation loss diverged sharply after epoch 3, rising to ~2.0 by epoch 9, with val accuracy collapsing from ~90% to ~75%, yielding only 91.26% test accuracy.

Global adaptive finetuning was the weakest strategy for both models. ResNet-50 selected 94.3% of all parameters as "important" effectively fine-tuning the entire network with a meaningless threshold yet achieved only 85.52% test accuracy in 256.3s. DenseNet-121 global selection achieved 84.59% in 666.8s. Both results are substantially below their standard fine-tuning counterparts.

4. Comparative Analysis

4.1 Architecture and Feature Reuse

ResNet-50's additive skip connection ($H(x) = F(x) + x$) does not explicitly accumulate features each residual block models only the change from the previous representation. Features from early layers are not directly visible to later layers unless re-activated through gradient flow. This means adaptive fine-tuning must span multiple layer groups to be effective; restricting updates to layer4 alone creates an information bottleneck explaining the ResNet last-block overfitting.

DenseNet-121's concatenative connectivity ($x_l = H([x_0,...,x_{l-1}])$) means denseblock4 already receives feature maps from every preceding layer. A filter that fires strongly in denseblock4 genuinely encodes a high-level, multi-scale representation relevant to CalTech-101. This is why last-block adaptive finetuning works well for DenseNet but poorly for ResNet the same strategy operates on fundamentally different information content in the two architectures.

4.2 Parameter Efficiency and Complexity

DenseNet-121 achieves comparable or better accuracy with 3.4x fewer total parameters (7.06M vs 23.7M) and 29% lower inference cost (2.9 vs 4.1 GFLOPs). In the adaptive last-block setting, DenseNet trains only 1.87M parameters (26.4%) to reach 92.68% test accuracy the best accuracy-per-parameter ratio across all experiments. Despite fewer parameters, DenseNet trains slower per experiment due to dense feature map caching increasing memory bandwidth requirements: Standard C2 DenseNet took 333s vs 147s for ResNet C2 despite similar per-epoch time (~11s).

4.3 Overfitting and Generalisation

ResNet-50 is more sensitive to the number of unfrozen layers. Case 1 (64% trainable) generalises well; Case 2 (93.9% trainable) overfits clearly on 6,400 training images. The global adaptive method, also at 94.3% trainable, similarly fails. The pattern is consistent: ResNet overfits when too many independent parameters are updated simultaneously.

DenseNet-121 is more robust to unfreezing depth. Dense connectivity acts as implicit regularisation gradient updates are distributed across shared concatenated features, reducing the effective degrees of freedom per parameter. DenseNet C2 (39.5% trainable) and last-block adaptive (26.4% trainable) both show stable, non-divergent validation curves.

4.4 Why DenseNet Fits CalTech-101 Better

CalTech-101 contains 102 visually diverse object categories (faces, vehicles, animals, instruments) with only ~60-80 images per class and significant variation in texture, scale, and background. DenseNet's explicit multi-scale feature reuse is well-aligned with this diversity low-level texture and edge features from early layers remain directly accessible at every subsequent layer, aiding discrimination of fine-grained visual differences. ResNet remains highly competitive (best speed-accuracy trade-off with Standard C1), but its independence between layer groups makes it more vulnerable to overfitting when the training set is small and diverse.

5. Discussion

The primary finding is that standard fine-tuning outperforms adaptive fine-tuning across all configurations. For CalTech-101's domain shift from ImageNet which is substantial given the dataset's diverse, non-photographic categories coherent layer-group adaptation is more effective than scattered filter-level updates. The adaptive methods' assumption that activation magnitude reliably proxies task relevance holds better for DenseNet (where denseblock4 aggregates the full feature hierarchy) than for ResNet (where layer4 is more isolated).

Global adaptive finetuning fails for a specific architectural reason: ResNet's activation magnitudes are distributed nearly uniformly across residual blocks, making the top-25% global threshold permissive to the point of selecting 94% of all filters. For DenseNet, early dense layers produce inflated activation magnitudes due to channel accumulation from concatenation contaminating the global importance ranking with low-level, already-optimal ImageNet filters.

One acknowledged limitation is CalTech-101's class imbalance. Without explicit class weighting, macro F1 scores should be interpreted cautiously they provide an unweighted average across 102 classes and may mask poor performance on rare categories. Future work could address this through loss re-weighting or class-balanced sampling.

6. Conclusion

This study demonstrates that transfer learning from ImageNet provides strong initialisation for both architectures on CalTech-101. Standard fine-tuning with carefully selected layer depth outperforms adaptive filter-importance methods overall. ResNet-50 Case 1 (fc + layer4, Adam lr=1e-4) offers the best speed-accuracy trade-off: 93.28% test accuracy in 100.9s. DenseNet-121 Case 2 (cls + denseblock4 + transition3, Adam lr=1e-3) achieves the best test accuracy (93.61%) and most stable convergence, at the cost of 3.3x longer training time. DenseNet's dense feature concatenation, lower parameter count, and robustness to selective fine-tuning make it the preferred architecture for small, visually diverse datasets such as CalTech-101.